

## 6.5 Types of Prototypes (Low, Mid, High-Fidelity)

فِذِلِیٹی (Fidelity) کا مطلب ہے کہ پروٹوٹائپ حتمی پراڈکٹ سے کتنا ملتا جلتا ہے۔

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 06 of 16

### 6.5 Types of Prototypes (Low, Mid, High-Fidelity)

#### LEARN IT

Prototypes are commonly divided into **three types** according to their level of detail, called **fidelity**.

**Fidelity means how close the prototype is to the final product.** Low fidelity = far from final; high fidelity = very close to final.

#### 6.5.1 Low-Fidelity Prototype

A **quick and rough** model made at the **very beginning**. It is **drawn on paper**. It does **not focus on colours or fonts**. It shows the **main idea and flow** (which screen comes after which). It is **very suitable for students** because it is **cheap, fast, and easy to change**.

#### 6.5.2 Mid-Fidelity Prototype

**Clearer than a paper sketch**. It is created in **PowerPoint, Canva, or Figma**. It shows screens and **navigation** in a more organised way, but **without final graphics**. It is useful to **test screen-to-screen navigation**, for example by linking slides so that clicking "Next" opens the next screen.

#### 6.5.3 High-Fidelity Prototype

Looks **very close to the final product**. It includes **realistic colours, buttons, sample data, and working features**. It **takes more time** and should **only be created after the basic idea is tested**.

#### Comparison of prototype types

For most school and college projects, you should **begin with a low-fidelity prototype**. A common mistake is to start building a full app before knowing whether users need it. A rough prototype protects you from this mistake.

**Table 6.3: Types of prototypes**

| Type          | Meaning                                      | Best Use                      | Example                                       |
|---------------|----------------------------------------------|-------------------------------|-----------------------------------------------|
| Low-Fidelity  | A quick and rough model of the idea.         | Very early testing.           | Hand-drawn app screens for homework reminder. |
| Mid-Fidelity  | A clearer model that shows layout and flow.  | Testing navigation.           | Clickable screens in PowerPoint or Figma.     |
| High-Fidelity | A detailed model close to the final product. | Testing realistic experience. | Working demo of a small order form website.   |

Low-Fi (Paper)



Mid-Fi (PowerPoint)



High-Fi (Figma/HTML)

Extra comparison to help you revise:

| Feature           | Low-Fidelity                 | Mid-Fidelity             | High-Fidelity                       |
|-------------------|------------------------------|--------------------------|-------------------------------------|
| Tool              | Paper and pencil             | PowerPoint, Canva, Figma | Figma (detailed), HTML              |
| Colours and fonts | No                           | Not final                | Realistic                           |
| Clickable?        | No (you move papers by hand) | Yes, basic links         | Yes, with working features          |
| Cost and time     | Lowest, fastest              | Medium                   | Highest, slowest                    |
| Easy to change?   | Very easy                    | Fairly easy              | Harder                              |
| When              | Very beginning               | After the flow is agreed | Only after the basic idea is tested |

#### WORKED EXAMPLE

**Worked example: one idea, three fidelities.** Idea: a Homework Reminder for a college in Jacobabad.

1. **Low-fi:** Sana draws three screens on paper in 15 minutes: "Today", "Add homework", "Reminder". Classmates point at the drawings to "use" them.
2. **Mid-fi:** After feedback, she makes three PowerPoint slides with grey boxes and links them with hyperlinks, so tapping "Add" opens the Add slide. Now she tests whether people can move between screens easily.
3. **High-fi:** Once users clearly want it, she builds a small HTML page with real colours, a working form and sample homework data.

Each step costs more, so she only moves up when the previous step proves the idea is worth it.

#### REMEMBER IT

**Fidelity = "faithfulness" to the final product.** Remember **Paper → PowerPoint → Polished** (low → mid → high).

#### COMMON MISTAKE

**Mistake:** Starting with a high-fidelity prototype or a full app because it "looks professional".

**Fix:** Begin with low fidelity. If users do not need the idea, you have lost only paper and an hour, not weeks of work.

#### اردو میں سمجھیں

فیدیلیٹی (Fidelity) کا مطلب ہے کہ پروٹوٹائپ حتمی پراڈکٹ سے کتنا ملتا جلتا ہے۔ لو فیدیلیٹی کاغذ پر بنایا گیا سستا اور جلدی بننے والا خاکہ ہے؛ مڈ فیدیلیٹی پاورپوائنٹ، کینوا یا فگما میں بنتا ہے اور اسکرینوں کے درمیان آنا جانا (Navigation) دکھاتا ہے؛ ہائی فیدیلیٹی بالکل اصل پراڈکٹ جیسا لگتا ہے اور اسے بنیادی خیال آزمانے کے بعد ہی بنانا چاہیے۔

#### CHECK YOURSELF

1. What does "fidelity" mean?
2. Which type of prototype is best for testing navigation?
3. Why should students begin with low fidelity?

Answers: 1. How close the prototype is to the final product. 2. Mid-fidelity (e.g. clickable screens in PowerPoint or Figma). 3. It is cheap, fast and easy to change, and it protects them from building a full app before knowing whether users need it.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

**MCQ 2.** VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

**MCQ 3.** The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

**MCQ 4.** Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

**MCQ 5.** A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

**MCQ 6.** Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

## Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

## Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Integrated Development Environment: editor + run + debug + files in one place.
2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

### Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.